MIL-R-12540B(EL)
6 February 1967
SUPERSEDING
MIL-R-12540A(SigC)
15 April 1958

MILITARY SPECIFICATION

RECTIFIER, METALLIC RA-91-( )

1. SCOPE

1.1 **Scope.** This specification covers one type full wave, metallic type, rectifier designed to operate from a 105 to 125 volts or 210 to 230 volts 50 to 60 cycles, single phase, alternating current, power source; and to furnish 6 to 60 volts, up to 12 amperes, for charging 4 Batteries BB-46. (24 lead acid battery cells in series.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871 — Twine, Cotton, Wrapping.

HH-I-595 — Insulation Tape, Electrical, Pressure-Sensitive Adhesive, Plastic, For Low-Temperature Application.

QQ-S-571 — Solder, Tin Alloy, Lead Tin Alloy and Lead Alloy.

PPP-B-636 — Box, Fiberboard.

PPP-B-640 — Box, Fiberboard, Corrugated, Triple-Wall.

PPP-F-320 — Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.

PPP-P-291 — Paperboard, Wrapping, Cushioning.

PPP-S-760 — Strapping, Nonmetallic (And Connectors).

PPP-T-45 — Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing.

PPP-T-76 — Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing).

PPP-T 97 — Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116 — Preservation, Methods of.

FSC 6130

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document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.3.2 *Units.* After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.4 **Finish.** The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings (see 4.4).

3.5 **Marking.**

3.5.1 *General.* Marking shall conform to MIL-M-13231 (see 4.4). Front panel marking shall be group I as described in that specification.

3.5.2 *Visibility.* Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 *Serial numbers.* Each Rectifier RA-91-( ) shall be serial numbered.

3.6 **Soldering.**

3.6.1 *Solder.* Solder shall conform to Specification QQ-S-571. For soldering electrical connections, composition SN60, type AR or S, shall be used for general purposes; and composition SN62 or SN63, type AR or S, for special purposes. When type S solder is used for soldering electrical connections, the flux shall conform to MIL-F-14256.

3.6.2 *Acid or acid salts.* No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.6.3 *Process.* There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.7 **Tropicalization.**

3.7.1 *Treating materials.* Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall detemine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.7.2 *Toxicity.* Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.7.3 *Flexibility.* Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected tc specified service conditions.

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clamping between a metallic and a nonmetallic material. Welded electrical connections shall not be used where it may be necessary to disconnect and reconnect the circuit during servicing. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to MIL-V-173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections, except during overall tropicalization of the equipment.

3.9 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment (see 4.15). Individual items shall not be hand-picked for fit or performance; however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.10 Input. The rectifier shall operate from an input of 105 to 125 volts or 210 to 230 volts, 50 to 60 cycles, single phase, alternating current power source.

3.11 Charging When operating from either input specified in 3.10, the rectifier shall be capable of furnishing direct current power, for charging 4 Batteries BB-46. (24 fully discharged lead-acid cells in series). The output circuit switch shall permit selection of charging current at 2, 4, 6, 8, 10 and 12 amperes. (See 4.6).

3.13 Ripple voltage. Ripple voltage shall not exceed ½ of 1 percent. (See 4.8).

3.13 Service conditions. The equipment shall meet the following service conditions—where a test is referenced, meeting the test shall be considered as compliance with the requirement:

3.13.1 *Temperature* (see 4.9).

(a) *Operating.* Ambient temperature in the range of +125°F to −40°F. (The 125°F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) *Nonoperating.* Exposure in the range of +160°F to −80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.13.2 *Relative humidity* (see 4.10). Up to 97 percent humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

3.13.3 *Elevation* (see 4.11).

(a) *Operating.* Up to 10,000 feet above sea level.

(b) *Nonoperating.* Up to 25,000 feet above sea level.

3.14 Vibration and shock.

3.14.1 *Vibration, internal.* With the equipment shock-mounts (if any) blocked, the amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second (see 4.12).

3.14.2 *Shock; bench handling.* The equipment shall meet the requirements of this specification when tested in accordance with 4.13.

3.15 Efficiency. The efficiency of Rectifier RA-91-( ) shall be not less than 70 percent when tested in accordance with paragraph 4.14.

3.16 Continuity. When tested as specified in paragraph 4.7, the output voltage shall increase progressively.

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the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	*Where required*
Finish	3.4
Marking	3.5

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government—at its discretion—may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements (see 6.4). Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 *Group A inspection.* This inspection including sampling, shall conform to table II and the inspection procedures of MIL-STD-105, using the general inspection levels. Group A inspection shall be performed in any order which is satisfactory to the Government.

TABLE II. *Group A Inspection*

Inspection	Rqt. para	Insp. para	AQL Major	AQL Minor
Visual and mechanical	3.18	4.16	1.5dphu	6.5dphu
Electrical:				
Ripple	3.12	4.8)	1.5dphu	
Efficiency	3.15	4.14)		*

* All electrical defects are considered Major.

4.5.2 *Group B inspection.* This inspection, including sampling, shall conform to table III and to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S-4 for normal, tightened, and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.5.2.1 *Order of inspection within group B.* Group B inspection shall be performed in any order which is satisfactory to the Government.

TABLE III. *Group B Inspection*

Inspection	Rqt. para	Insp. para
Interchangeability	3.9	4.15
Continuity test	3.16	4.7

4.5.3 *Group C inspection.* This inspection shall consist of the tests specified in table IV and shall be performed on sample units that have been subjected to and met groups A and B inspection. Sample units shall be selected in accordance with paragraph 4.5.3.1.

TABLE IV. *Group C Inspection*

Inspection	Rqt. para	Insp. para
Vibration	3.14.1	4.12
Shock; bench handling	3.14.2	4.13
Moisture resistance	3.13.2	4.10
Charging	3.11	4.6
Temperature	3.13.1	4.9
Elevation	3.13.3	4.11

4.5.3.1 *Sampling for inspection.* One sample unit shall be selected at random for all

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4.9 Temperature test. The equipment shall be subjected to the temperature cycle shown on MIL-STD-169. Measurements shall be taken at +125°F step 4; −40°F, step 8 and +77°F, step 10. The equipment shall meet the requirements of this specification.

4.10 Moisture-resistance test for equipment. The equipment shall be tested as follows:

4.10.1 *Test conditions.*

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Test sealed equipment with the seal intact except during last two cycles. Bring out leads for electrical measurements through a hole in the equipment enclosure and seal against entrance of moisture. Perform last two cycles with the chassis removed from its enclosure.

4.10.2 *Test method.*

(a) Dry at 130° ± 5°F for 24 hours.

(b) Condition at 77° ± 5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to MIL-STD-170.

(d) After cycling has been completed, condition the equipment for 24 hours at 77° ± 5°F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specification performance.

4.11 Altitude test. The equipment shall be placed within an altitude chamber and the internal pressure reduced to 20.58 inches of mercury (corresponding to an altitude of 10,000 feet above sea level). The minimum duration of this test period shall be one hour. During this period the equipment shall be tested to determine conformance with all the electrical operating requirements of this specification.

4.12 Vibration test. The equipment shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.14.1, as follows:

(a) The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. Equipment shock mounts, if any, shall be blocked. The vibration table shall provide approximately sinusoidal vibration (see 4.12(d)).

(b) The equipment shall be vibrated successively in three mutual perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per second steps. The total excursion shall be constant at 1/64 inch.

(c) Mechanical resonance, if any, of the complete structure, of subassemblies, and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts.

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freedom from operational malfunction caused by the rough handling:

Visual and mechanical inspection	4.16
Continuity test	4.7

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2(c)).

5.1.1 *Level A.*

5.1.1.1 *Cleaning.* Rectifier RA-91-( ), shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 *Drying.* Drying shall be accomplished in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 *Unit packaging.* Unit packaging shall be in accordance with the applicable procedures of the methods prescribed in MIL-P-116 and as specified herein.

5.1.1.3.1 *Technical manual.* Each technical manual shall be packaged method IC-1.

5.1.1.3.2 *Lamps.* Package each lamp method III, as follows: cushion each lamp by wrapping in paperboard conforming to PPP-P-291, type III. Secure the wrap with gummed paper tape conforming to PPP-T-45 type III.

5.1.1.3.3 *Rectifier RA-91-( ).* Each rectifier shall be packaged method III, as follows: wind each cord into a coil of proportionate dimensions and tie at three places with cotton twine conforming to T-T-871. Secure the plugs to the coils with the ties. Secure the coiled cords and the lamps packaged as specified in 5.1.1.3.2 to rectifier with water-resistant pressure-sensitive tape, conforming to PPP-T-76. Cushion each rectifier on all surfaces with cells or pads, or both, fabricated of fiberboard conforming to PPP-F-320, grade W6c, designed to absorb the shock of impact in handling and transit. Place the cushioned rectifier within a close fitting fiberboard box conforming to PPP-B-636, grade W5c. Place the technical literature, packaged as specified in 5.1.1.3.1 on top of the rectifier directly under the lid of the box. Close the box in accordance with the appendix of the box specification. In addition, seal all seams and joints with tape, not less than two inches wide, conforming to PPP-T-76.

5.1.2 *Level C.* Rectifier RA-91-( ) shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2(c)).

5.2.1 *Level A.*

5.2.1.1 *Palletized load.* Rectifier's packaged as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather resistant, W5c. The load shall be bonded to the pallet by strapping.

5.2.1.2 *Less than pallet load.* When quantities per destination are less than a pallet load, rectifiers shall be packed within fiberboard boxes conforming to PPP-B-640, Class 2, grade A, style E, or MIL-B-43291. Closure shall be in accordance with the applicable box specification. Reinforcing shall be made by nonmetallic strapping or pressure sensitive tape, conforming to PPP-S-760, type II or PPP-T-97 type IV, respectively, selection of the material and application shall be in accordance with the applicable box specification.

5.2.2 *Level B.* Rectifier RA-91-( ), packaged as specified in 5.1, shall be packed as

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEM PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1426 APR 63

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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